

```

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler, StandardScaler

# Load Dataset (required for visualizations)
df = pd.read_csv('/content/boston.csv').head(15)

# Perform transformations needed for visualizations
df['Log_MEDV'] = np.log1p(df['MEDV'])

scaler_minmax = MinMaxScaler()
df['AGE_MinMax'] = scaler_minmax.fit_transform(df[['AGE']])

scaler_std = StandardScaler()
df['AGE_Standard'] = scaler_std.fit_transform(df[['AGE']])

bins = [0, 25, 50, 100]
labels = ['Young', 'Adult', 'Senior']
df['AGE_Group'] = pd.cut(df['AGE'], bins=bins, labels=labels)

skewed_data = np.array([1, 2, 2, 3, 100])
log_fixed = np.log(skewed_data)

# -----
# 2. TRANSFORMATION (Log) - Visualizations
# -----
plt.figure()
plt.hist(df['MEDV'])
plt.title("Original MEDV Distribution")
plt.xlabel("MEDV")
plt.ylabel("Frequency")
plt.show()

plt.figure()
plt.hist(df['Log_MEDV'])
plt.title("Log Transformed MEDV")
plt.xlabel("Log MEDV")
plt.ylabel("Frequency")
plt.show()

# -----
# 4. BINNING - Visualizations
# -----
plt.figure()
df['AGE_Group'].value_counts().plot(kind='bar')
plt.title("AGE Group Distribution")
plt.show()

# -----
# 5. FIXING SKEWED DATA - Visualizations
# -----
plt.figure()
plt.hist(skewed_data)
plt.title("Skewed Data")
plt.show()

plt.figure()
plt.hist(log_fixed)
plt.title("After Fix (Log Transformation)")
plt.show()

```

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FileNotFoundError                                Traceback (most recent call last)
/tmp/ipykernel_1091/311907805.py in <cell line: 0>()
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler, StandardScaler

# Load Dataset (using first 15 rows as previously requested)
df = pd.read_csv('/content/boston.csv').head(15)

# Perform scaling transformations for visualizations
scaler_minmax = MinMaxScaler()
df['AGE_MinMax'] = scaler_minmax.fit_transform(df[['AGE']])

scaler_std = StandardScaler()
df['AGE_Standard'] = scaler_std.fit_transform(df[['AGE']])

plt.figure(figsize=(10, 6))
plt.plot(df['AGE'], label='Original AGE', marker='o')
plt.plot(df['AGE_MinMax'], label='Min-Max Scaled AGE', marker='x')
plt.plot(df['AGE_Standard'], label='Standardized AGE', marker='s')
plt.title('Comparison of Original, Min-Max Scaled, and Standardized AGE')
plt.xlabel('Index')
plt.ylabel('Value')
plt.legend()
plt.grid(True)
plt.show()
```

